Torsional Analysis of an Exoskeleton Motor-Cage Output Joint

Bolt-Slip Prediction and Gusset-Bracket Validation with Preloaded-Bolt FEA

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Abstract

The output side of an upper-limb exoskeleton joint transmits the torque of a CubeMars AKH70-16 actuator (rated 26 N m, peak 78 N m) from the rotor flange to the next link through a 4×M4 bolted joint between two 12.7 mm (0.5 in) 6061-T6 plates. A bolt-group hand calculation showed that, at peak torque, the required shear on the outermost bolt (1 494 N) exceeds the friction capacity of a preloaded M4 joint (593 N), so the joint would slip. Two L-shaped gusset brackets were added. A SolidWorks Simulation static study with 12 preloaded bolt connectors, frictional contact and a three-level mesh-convergence check shows that the brackets reduce the shear on the original bolts by a factor of 32, keep every joint below the slip threshold, and increase the torsional stiffness of the output joint by a factor of 8.9. The highest converged stress in the bracket is 55 MPa (static safety factor 5.0); the highest converged stress in the assembly, 208 MPa, is preload bearing at a bolt-hole edge and motivates the use of washers.

1 Objective and scope

The motor cage holds the AKH70-16 stator on the input side and carries its rotor output to the next link on the output side. The output load path studied here is

rotor flange → Rotating Spacer (⌀50×15) → Moving Side Plate (12.7 mm) → 4×M4 → Static Connector (12.7 mm) → shaft support,

with two 5 mm Bracket gussets bolted across the corner between the Moving Side Plate and the Static Connector as a parallel path. The analysis answers two questions:

1. Without the brackets, does the 4×M4 joint slip at peak torque?
2. With the brackets, how far is the bolt shear reduced, how much stiffer is the joint, and how highly stressed are the brackets themselves?

The input (stator) side, the actuator internals and the shaft supports are outside the scope of this report.

2 Hand calculations

2.1 Bolt preload and friction capacity

For an M4 class 8.8 screw (tensile stress area $A_s = 8.78\,\mathrm{mm}^2$, proof strength $S_p = 600\,\mathrm{MPa}$) tightened to 75 % of proof,

$$F_i = 0.75\,S_p A_s = 0.75(600)(8.78) \approx 3\,950\,\mathrm{N}, \tag{1}$$

which with a nut factor $K = 0.2$ corresponds to a tightening torque

$$T_{\mathrm{tighten}} = K F_i d = 0.2(3950)(0.004) \approx 3.2\,\mathrm{N\,m}. \tag{2}$$

With a single faying surface and $\mu = 0.15$, the shear a bolt can transmit by friction before slipping is

$$F_{\mathrm{slip}} = \mu F_i = 0.15(3950) \approx 593\,\mathrm{N}. \tag{3}$$

2.2 Bolt-group shear

The 4×M4 joint is a bolt group loaded by a pure in-plane moment equal to the motor torque T. Assuming the plates are rigid and all of T is carried by bolt shear, the force on bolt i at distance r_i from the group centroid is

$$F_i = \frac{T\,r_i}{\sum_j r_j^2}. \tag{4}$$

The four bolts lie on one line at $x = \pm 9\,\mathrm{mm}$ and $\pm 22.5\,\mathrm{mm}$, so $\sum r^2 = 2(9^2 + 22.5^2) = 1\,174.5\,\mathrm{mm}^2$. Adding the four bracket screws on the plate edges (at 38.1 mm and 57.0 mm from the centroid) and treating the brackets as rigid raises $\sum r^2$ to about $10\,576\,\mathrm{mm}^2$. Figure 1 shows the geometry and Table 1 the results.

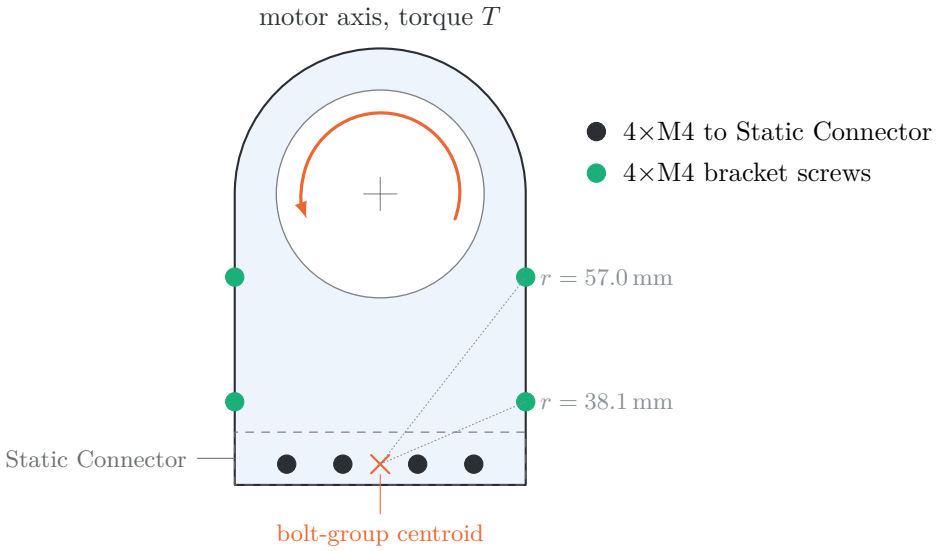


Figure 1: Moving Side Plate (top view, plate coordinates). The 4×M4 joint is a single row spanning only 45 mm; the bracket screws add load points far from the centroid.

Table 1: Hand-calculation results.

Quantity	Peak, 78 N m	Rated, 26 N m
Rotor 6×M4 on PCD 35, shear per bolt	743	248
4×M4, outer bolt shear, no bracket	1 494	498
4×M4, outer bolt shear, rigid bracket	166	55
Bracket screw, maximum shear, rigid bracket	420	140
Friction capacity per M4, F_{slip}	593	

All forces in newtons.

Without the brackets the outer bolt needs 1 494 N against a capacity of 593 N; the joint is predicted to slip at peak torque and to be marginal (498 N) even at rated torque. With rigid brackets every bolt stays well below the slip threshold.

2.3 Bearing under the screw head

The preload acts on the annulus between the 7 mm head and the 4.3 mm hole:

$$p = \frac{F_i}{\frac{\pi}{4}(7^2 - 4.3^2)} = \frac{3950}{24.0\,\text{mm}^2} \approx 165\,\text{MPa}, \tag{5}$$

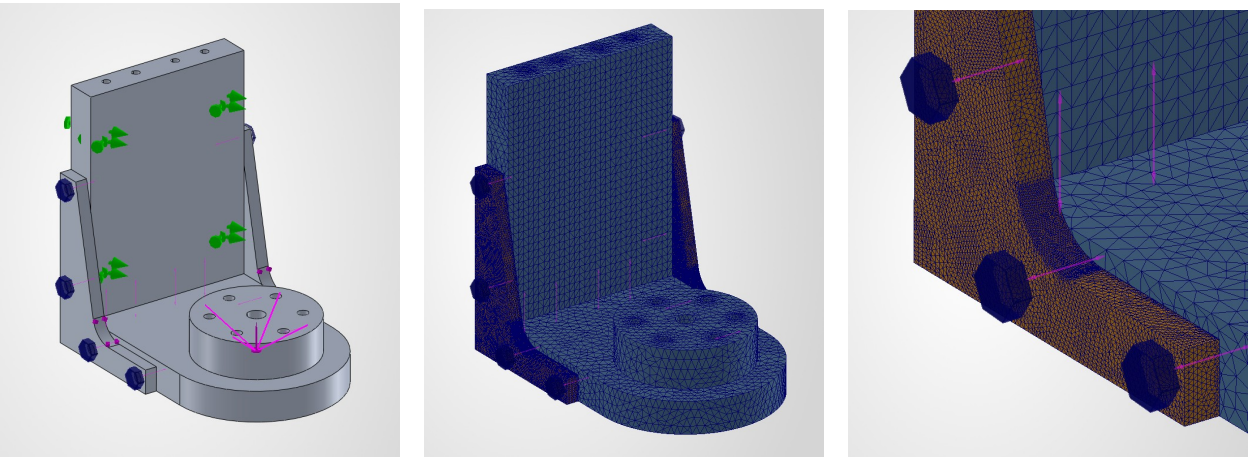
below the bearing yield strength of 6061-T6 (about 386 MPa) but high enough that embedding and preload loss are a concern for an aluminium joint under cyclic load.

3 Finite-element model

Table 2 summarises the study, which was built in SOLIDWORKS Simulation (Premium for Students 2026 SP4.1) on a dedicated configuration in which the actuator, the input-side plates, the Moving Connector, the shaft supports and the spacers are suppressed. Figure 2 shows the boundary conditions and mesh.

Table 2: Simulation set-up.

Item	Setting
Study type	Linear static, FFEPlus solver, small displacement
Bodies	Rotating Spacer, Moving Side Plate, Static Connector, Bracket 1, Bracket 2
Material	6061-T6: $E = 69\,\text{GPa}$, $\nu = 0.33$, $S_y = 275\,\text{MPa}$
Global interaction	Contact (no penetration), $\mu = 0.15$
Local interaction	Rotating Spacer $\leftrightarrow$ Moving Side Plate bonded (clamped by the rotor screws; not the joint under study)
Bolt connectors	12 $\times$ "standard or counterbore screw": 4 plate-to-connector, 8 bracket screws. Head 7 mm, shank 4 mm, alloy steel, distributed coupling, axial preload 3 950 N
Fixture	Fixed geometry on the four M6 tapped-hole faces of the Static Connector (attachment to the shaft support)
Load	Remote load, rigid connection, on the rotor-side face of the Rotating Spacer; moment 78 N m about the motor axis
Mesh	Curvature-based, second-order tetrahedra; see Table 3



(a) Fixtures (green), torque (magenta) and bolt connectors.

(b) Mesh, fine2 level.

(c) Local refinement at the bracket R10 fillet.

Figure 2: Model, boundary conditions and mesh.

4 Mesh convergence

Three meshes were run with brackets fitted. *fine* refines the two side faces of the Moving Side Plate that the brackets bear on; *fine2* additionally refines the brackets. Table 3 lists the quantities used in the conclusions.

Table 3: Convergence of the with-bracket study (peak torque).

	coarse	fine	fine2	Δ fine→fine2
Elements	250 765	529 313	846 763	
Global max / min size (mm)	3 / 0.5	3 / 0.5	3 / 0.3	
Mesh controls (mm)[a]	0.6 / 1.5 / –	0.6 / 1.5 / 0.4	0.3 / 0.75 / 0.4	
Max displacement (mm)	0.09318	0.09303	0.09296	−0.1%
R10 fillet, mean (MPa)	22.78	23.43	23.10	−1.4%
R10 fillet, max (MPa)	55.33	55.43	55.37	−0.1%
Bolt-hole edge, mean (MPa)[b]	138.0	178.7	177.3	−0.8%
Bolt-hole edge, max (MPa)[b]	219.8	206.2	208.0	+0.9%
Max bolt shear (N)[c]	119.8	–	119.4	−0.3%
Global max von Mises (MPa)	219.8	230.5	237.6	+3.1%, singular

[a] R10 fillet faces / bracket bodies / plate side faces.
[b] Bracket 2, hole (50, 5), edge on the face clamped against the Moving Side Plate.
[c] Bolt forces were exported for coarse and fine2 only; Δ is coarse→fine2.

Displacement, bolt shear and the bracket-fillet stress converge within 1.5 %. The bolt-hole edge changes by 30 % from coarse to fine because its *contact partner*, the side face of the plate, is refined; once both sides of the interface are fine it changes by less than 1 % (Figure 3). The global maximum keeps rising and moves location between meshes; it sits on bolt-connector and contact-edge singularities and is not used for design.

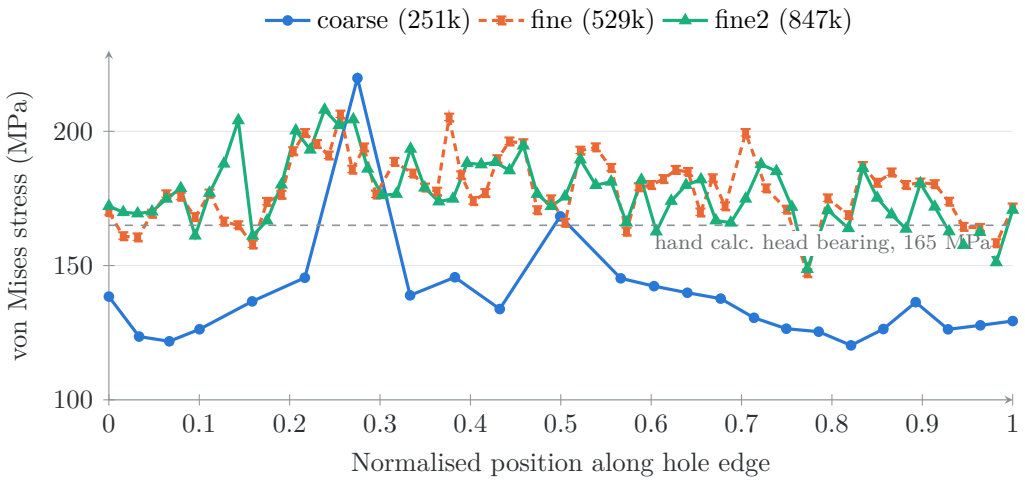


Figure 3: von Mises stress around the clamped edge of bracket hole (50, 5) for the three meshes. The coarse curve has single-node spikes; fine and fine2 overlap.

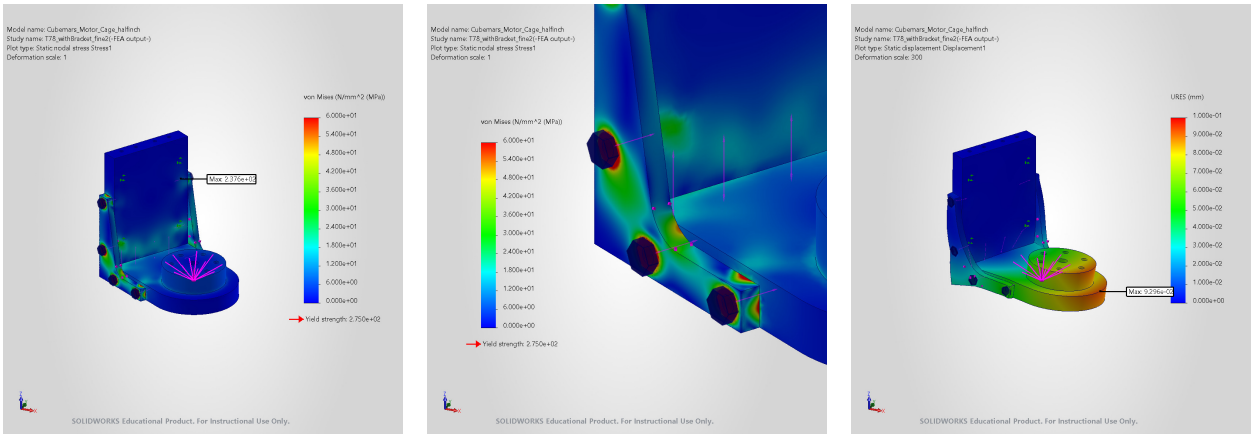
5 Results with the brackets

5.1 Bolt forces

All 12 connectors hold their preload (3 917 N–3 976 N). The largest shear on any bolt is 119 N, a fifth of the friction capacity, so no joint slips at peak torque. The four plate-to-connector bolts carry 6 N–25 N (Table 4); their bending moments are below 0.25 N m.

The FEA shear is far below the rigid-bracket hand estimate because the hand calculation routes all torque through bolt shear. In the model, preload clamps the faying surfaces and most of the torque is carried by friction and contact pressure; bolt shanks see only the small elastic mismatch. This is the intended behaviour of a preloaded joint.

5.2 Stress and displacement



(a) von Mises, 0 MPa–60 MPa scale. **(b)** Bracket corner and R10 fillet. **(c)** Resultant displacement, 300×.

Figure 4: With brackets, fine2 mesh, peak torque.

- **Bracket fillet.** Mean 23 MPa; maximum 55 MPa at the inner face where the plate corner, the Static Connector end face and the bracket meet. The maximum is mesh-independent (Table 3), so it is treated as a real local concentration: static safety factor $275/55 = 5.0$, and $96/55 \approx 1.7$ against the fatigue strength of 6061-T6 at 5×10^8 cycles for fully reversed loading.
- **Bolt-hole edges.** 150 MPa–208 MPa around the clamped face of the bracket holes, consistent with the 165 MPa head-bearing estimate. This stress is set by preload, is compressive and barely changes with joint torque, so it is a static rather than a fatigue concern; local safety factor $275/208 \approx 1.3$.
- **Displacement.** Maximum 0.093 mm at the far edge of the plate.

6 Comparison: without the brackets

A copy of the coarse study with both brackets excluded and their eight connectors deleted isolates the effect of the brackets. Coarse-level results are adequate for this comparison because displacement and bolt shear converge within 0.4 % (Table 3).

6.1 Rotation angle

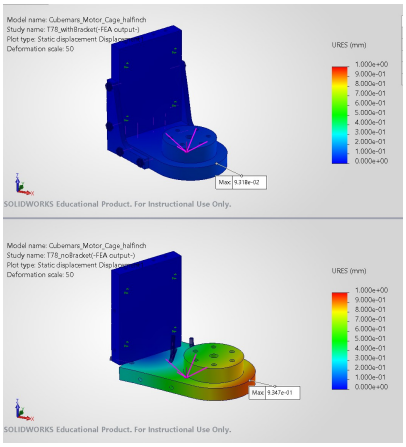
Without the brackets the plate rotates about the bolt row, not about the motor axis (Figure 5). The displacement ratio around the rim of the 50 mm spacer confirms a near-rigid rotation: $u_{\min}/u_{\max} = 0.375/0.849 = 0.44$, equal to the ratio of the rim's nearest and farthest distances from the bolt row,

$40/90 = 0.44$. Because the pivot is not on the motor axis, the rotation angle is taken from the displacement difference across the rim diameter D, which is independent of the pivot location:

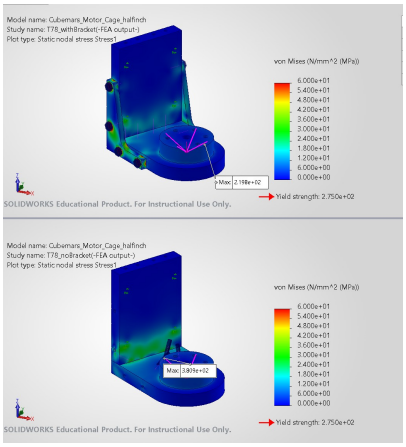
$$\theta \approx \frac{u_{\max} - u_{\min}}{D}, \qquad K_\theta = \frac{T}{\theta}. \tag{6}$$

Table 4: With versus without brackets, peak torque, coarse mesh.

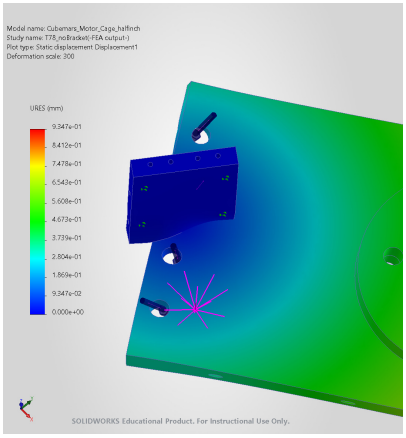
	With brackets	Without	Ratio
Screw 1, outer, shear (N)	24.8	768.6	×31
Screw 2, inner, shear (N)	7.6	283.6	×37
Screw 3, inner, shear (N)	6.0	295.8	×49
Screw 4, outer, shear (N)	20.6	782.7	×38
Max bolt bending moment (N m)	0.23	6.72	×29
Joint state	no slip	slip	
Max displacement (mm)	0.093	0.935	×10
Spacer rim $u_{\max}$ / $u_{\min}$ (mm)	0.0920 / 0.0387	0.849 / 0.375	
Rotation θ (deg)	0.061	0.543	×8.9
Torsional stiffness K_θ (kN m rad^{-1})	73.2	8.2	×8.9
Tip motion of a 300 mm link (mm)	0.32	2.8	



(a) Displacement, common 0 mm–1 mm scale, 50×; with (top) and without (bottom) brackets.



(b) von Mises, common 0 MPa–60 MPa scale; with (top) and without (bottom).



(c) Without brackets, 300×: the plate swings about the bolt row.

Figure 5: Effect of removing the brackets.

6.2 Interpretation

- The outer bolts carry 770 N–783 N, above the 593 N friction capacity: the joint slips, as the hand calculation predicted.
- The outer-to-inner shear ratio, $775/290 = 2.7$, is close to the bolt-group ratio $22.5/9 = 2.5$ from Equation (4), which supports the model.

- The bolt shear accounts for $2(775.7)(22.5) + 2(289.7)(9) \approx 40\,\mathrm{N\,m}$, about half of the applied torque; contact friction and bearing carry the rest. The hand calculation, which assigns all torque to bolt shear, is therefore conservative.
- The bolt connector has no shank-to-hole contact, so after slip the shank is loaded as a cantilever and the $6.7\,\mathrm{N\,m}$ bending moment and $0.93\,\mathrm{mm}$ displacement overstate a real joint. In hardware the plate would slide through the $0.2\,\mathrm{mm}$ radial hole clearance and then bear on the screws, producing backlash every time the torque reverses. For an exoskeleton joint this lost motion is the more serious consequence.

7 Limitations

- Linear static analysis at a single torque; friction is handled by the contact solver but there is no load history or slip hysteresis.
- The actuator is replaced by a rigid remote load, which stiffens the spacer face.
- Bolt connectors carry preload and shear through a beam idealisation with no thread or shank-to-hole contact; stresses within about one bolt diameter of a connector are not reliable.
- Fatigue is assessed only by comparing the converged peak stress with the fully reversed fatigue strength; preload mean stress is not combined.
- Plate flatness, hole position tolerance and the coplanarity of the two faces the brackets bear on are not modelled; they are addressed in the GD&T stack-up.

8 Conclusions and recommendations

1. The original 4×M4 joint slips at the actuator's peak torque. Hand calculation and FEA agree on this, and the FEA bolt-shear distribution matches bolt-group theory.
2. The gusset brackets keep every joint below the slip threshold with a margin of about five, cut the shear on the original bolts by a factor of 32 and raise the torsional stiffness of the output joint by a factor of 8.9.
3. The brackets are lightly stressed (converged maximum $55\,\mathrm{MPa}$, static safety factor 5.0) and could be lightened if mass becomes a priority.
4. The governing stress is preload bearing at the bracket-hole edges ($208\,\mathrm{MPa}$, local safety factor 1.3). Fit M4 flat washers (DIN 125, $9\,\mathrm{mm}$ OD) under every head, which roughly halves the bearing pressure, and consider reducing the tightening torque to $2.5\,\mathrm{N\,m}$–$2.8\,\mathrm{N\,m}$; the friction margin above allows this.

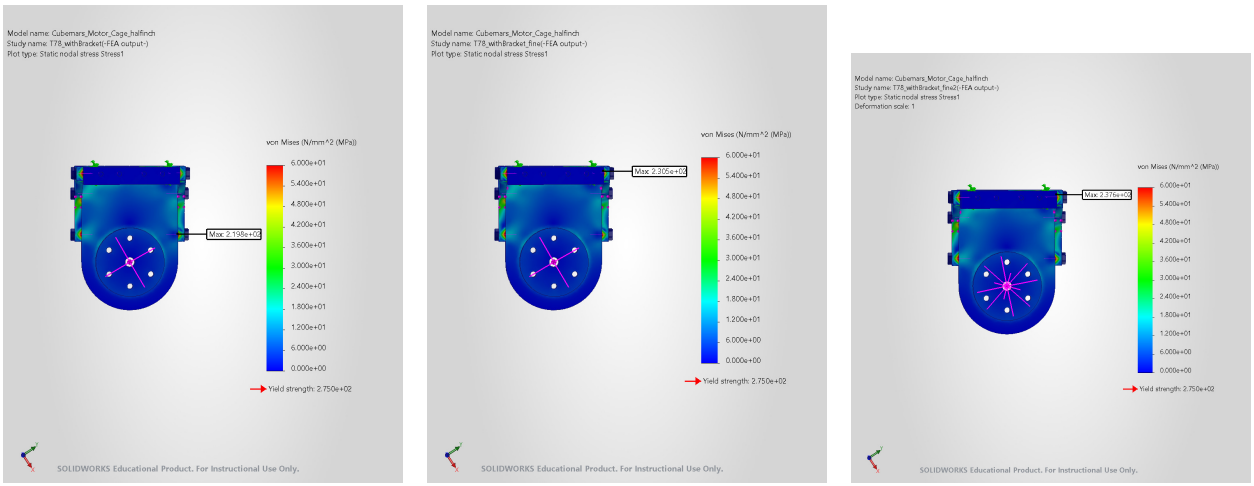
A Bolt connector forces, with brackets

Screws 1–4 are the plate-to-connector bolts (they are the four that remain in the without-bracket study); screws 5–12 are the bracket screws.

Table 5: Connector results, peak torque.

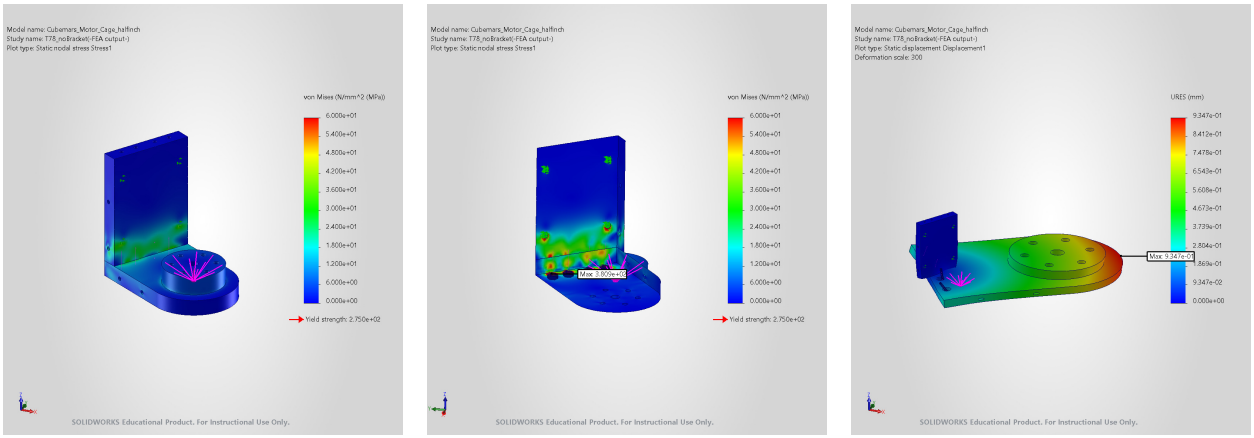
Connector	Shear, coarse (N)	Shear, fine2 (N)	Axial, fine2 (N)	Bending, fine2 (N m)
Screw 1	24.8	24.6	3 945	0.17
Screw 2	7.6	7.6	3 941	0.07
Screw 3	6.0	6.1	3 937	0.09
Screw 4	20.6	20.4	3 929	0.23
Screw 5	4.7	4.4	3 952	0.04
Screw 6	119.8	119.4	3 947	0.52
Screw 7	9.0	8.1	3 974	0.06
Screw 8	19.5	17.5	3 976	0.06
Screw 9	4.4	4.4	3 947	0.02
Screw 10	99.6	97.2	3 923	0.49
Screw 11	29.6	29.4	3 917	0.14
Screw 12	18.5	18.0	3 918	0.10

B Additional result plots



(a) Coarse, max 219.8 MPa. **(b)** Fine, max 230.5 MPa. **(c)** Fine2, max 237.6 MPa.

Figure 6: Location of the global stress maximum for the three meshes. It moves between singular points and keeps rising with refinement.



(a) Without brackets, von Mises, 0 MPa–60 MPa.

(b) Without brackets, global max 380.9 MPa at a bolt connector.

(c) Without brackets, displacement, 300×.

Figure 7: Without-bracket study, coarse mesh.